TEXORA · LECTURE SHEET

The Roving, Ring, Autocone and Rotor Spinning Frame

Four machines complete the spinning line: the roving frame attenuates sliver to a spinnable roving, the ring frame drafts, twists and winds yarn onto cops, the autocone rewinds cops into large, cleared cones, and the rotor frame spins directly from sliver. Every machine's delivery speed and production is calculated here from one relation — delivery = rotating speed ÷ twist.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 5 · SPECIALIZATION — YARN MANUFACTURING

1. The master relation

Wherever twist is inserted by a rotating element, the length delivered per minute is fixed by how many turns are put into each unit length:

$$\text{Delivery (per min)} = \text{rotating element speed (rpm)} / \text{twist (turns per unit length)}$$

$$\text{Production} = \text{delivery speed} \times \text{linear density of the delivered strand}$$

Twist itself is specified by the twist multiplier (TM):

$$\text{Twist (tpi)} = \text{TM} \times \sqrt{(\text{Ne})}$$

2. The roving (simplex) frame

Drawframe sliver at 3.84 g/m is still far too heavy to spin: the ring frame's drafting system works best with a draft of about 20–40. The roving frame applies a draft of about 8, inserts a small protective twist, and winds the roving onto bobbins that fit the ring-frame creel.

WORKED EXAMPLE — ROVING DRAFT, DELIVERY AND BOBBIN CONTENT

Feed sliver = 3.841 g/m (3841 tex); roving draft = 7.8.

Roving weight = 3841 / 7.8 = 492 tex = **0.492 g/m**, i.e. hank = 590.5/492 = **1.20 Ne**.

Twist: TM = 1.1, so TPI = 1.1 × √1.20 = 1.21 turns/inch.

Flyer speed = 1100 rpm → delivery = 1100 / 1.21 = 913 inch/min = **23.2 m/min**.

Production per spindle = 23.2 × 0.492 = 11.4 g/min = **0.69 kg/h**.

A bobbin holding 900 g of roving contains 900 / 0.492 = **1828 m**, and fills in 1828 / 23.2 = **79 minutes** — the doffing interval the creel schedule is built around.

3. The ring frame

The ring frame drafts the roving to final count, inserts the spinning twist with the spindle and traveller, and winds yarn onto a cop. Its delivery is set by the front roller through the twist relation applied at the spindle.

WORKED EXAMPLE — RING FRAME, NE 30

Roving = 492 tex; yarn = Ne 30 = 19.68 tex.

Ring draft = 492 / 19.68 = **25.0**.

Twist: TM = 3.9 → TPI = 3.9 × √30 = 21.4 turns/inch.

Spindle speed = 12,500 rpm → delivery = 12,500 / 21.4 = 585 inch/min = **14.9 m/min**.

Production per spindle = 14.9 × 0.01968 g/m = 0.293 g/min = **17.6 g/h**.

A frame of 1008 spindles therefore produces 1008 × 17.6 g/h ≈ **17.7 kg/h** at 100%.

Cop content: a cop of 55 g holds 55 / 0.01968 = **2794 m** of yarn, so doffing falls due every 2794 / 14.9 = **188 minutes** (about 3.1 h) per spindle position.

Note the scale: one ring spindle makes grams per hour where a drawframe delivery makes a hundred kilograms. Ring spinning is therefore always the most numerous machine in the mill, and its spindle count — not any other machine — sets mill capacity (Topic 8).

4. The autocone (winding)

Winding clears yarn faults at electronic clearers, splices the breaks, and builds the large cone the customer wants. There is no draft and no twist change; the calculations are package content and effective time.

WORKED EXAMPLE — CONE CONTENT AND WINDING TIME

Cone weight = 2.2 kg of Ne 30 yarn (0.01968 g/m).

Length on cone = 2200 / 0.01968 = **111,770 m** (about 112 km of yarn).

At a winding speed of 1250 m/min the pure winding time = 111,770 / 1250 = **89 minutes**.

Winding stops for clearer cuts, splicing and doffing; at a measured efficiency of 72% the real time = 89 / 0.72 = **124 minutes** per cone.

The same length figure gives clearer activity: at 90 cuts per 100 km, this cone's yarn will be cut about 0.9 × 111.8 ≈ 101 times — each cut a fault removed before it can reach a fabric.

5. The rotor frame

Rotor spinning opens sliver to single fibres with an opening roller, collects them in the groove of a high-speed rotor, and twists them into yarn in one operation — no roving frame, no separate winding package size limit, and delivery speeds an order above ring spinning. The price is a somewhat weaker, harsher yarn, best suited to coarse and medium counts.

WORKED EXAMPLE — ROTOR SPINNING, NE 20

Yarn = Ne 20 = 29.5 tex. Feed = drawframe sliver 3841 tex.

Rotor draft = 3841 / 29.5 = **130** (the opening roller does the attenuation work).

Twist: TM = 4.3 → TPI = 4.3 × √20 = 19.2 turns/inch = 757 turns/m.

Rotor speed = 90,000 rpm → delivery = 90,000 / 757 = **118.9 m/min** — eight times the ring frame's delivery.

Production per head = 118.9 × 0.02952 = 3.51 g/min = **211 g/h**, about twelve ring spindles' worth from one head.

A 3 kg package holds 3000 / 0.02952 = **101,609 m** and runs for 101,609 / 118.9 = 854 min ≈ **14.2 h** between doffs.

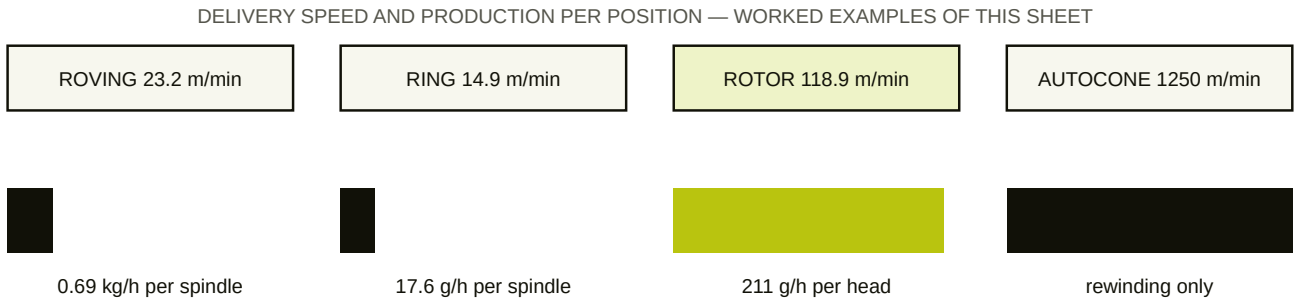


Figure 1 — The four machines compared on the numbers calculated above. Ring spinning's low delivery per spindle is why it dominates machine counts; rotor spinning's speed is why it dominates coarse-count economics.

6. Ring versus rotor — the calculation view

QUANTITY	RING (NE 30)	ROTOR (NE 20)
Feed	Roving 492 tex	Sliver 3841 tex
Draft	25	130
Twist insertion	Spindle/traveller, 12,500 rpm	Rotor, 90,000 rpm
Delivery	14.9 m/min	118.9 m/min

Production per position	17.6 g/h	211 g/h
Package	Cop 55 g (needs rewinding)	Cheese/cone up to 3 kg+
Processes eliminated	—	Roving frame and separate winding

Glossary

Twist multiplier (TM) — the constant linking twist to count: TPI = TM × √Ne. **Traveller** — the small clip on the ring that lags the spindle and so inserts twist and winds the cop. **Cop** — the yarn package built on the ring-frame spindle. **Clearer** — the device that detects and cuts out yarn faults during winding. **Splice** — a knotless join made by untwisting and re-twisting two yarn ends with air. **Opening roller** — the clothed roller that combs sliver into single fibres at the rotor machine. **Rotor groove** — the collecting surface inside the rotor where the fibre ring forms.

Key points

• Delivery = rotating speed / twist; production = delivery × linear density. Everything in this sheet follows from these two lines.
• Roving: draft 7.8 → 492 tex (1.20 Ne); delivery 23.2 m/min; 0.69 kg/h per spindle; bobbin doff ≈ 79 min.
• Ring Ne 30: draft 25, delivery 14.9 m/min, 17.6 g/h per spindle, cop doff ≈ 188 min.
• Autocone: a 2.2 kg cone holds ~112 km; 89 min pure winding, 124 min at 72% efficiency.
• Rotor Ne 20: draft 130, delivery 118.9 m/min, 211 g/h per head, package run ≈ 14.2 h.

Practice

1. A ring frame spins Ne 24 with TM 4.0 at a spindle speed of 13,000 rpm. Calculate TPI, delivery in m/min, and production per spindle in g/h. (Ne 24 = 24.6 tex.)
2. A rotor machine spins Ne 16 (36.9 tex) with a rotor speed of 100,000 rpm and TM 4.5. Calculate the twist in turns/m and the delivery speed.
3. A roving bobbin holds 750 g of roving of 0.55 g/m and the frame delivers at 22 m/min. Calculate the doffing interval.

Answer key

1. TPI = 4.0 × √24 = 4.0 × 4.90 = **19.6**. Delivery = 13,000 / 19.6 = 663 inch/min = **16.8 m/min**. Production = 16.8 × 0.0246 = 0.414 g/min = **24.9 g/h**.
2. TPI = 4.5 × √16 = 4.5 × 4 = 18.0 → turns/m = 18.0 × 39.37 = **709**. Delivery = 100,000 / 709 = **141 m/min**.

3. Length on bobbin = 750 / 0.55 = 1364 m. Interval = 1364 / 22 = **62 minutes**.
